elementary particles should correspond to the lowest-energy oscillatory patterns, the existence of multiple holes—somewhat like those in the multidoughnut—means that the patterns of string vibrations will fall into multiple families. If the curled-up Calabi-Yau has three holes, then we will find three families of elementary particles.16 And so, string theory proclaims that the family organization observed experimentally, rather than being some unexplainable feature of either random or divine origin, is a reflection of the number of holes in the geometrical shape comprising the extra dimensions! This is the kind of result that makes a physicist's heart skip a beat.

You might think that the number of holes in the curled-up Planck-sized dimensions—mountaintop physics par excellence—has now kicked an experimentally testable stone down to accessible energies. After all, experimentalists can establish—in fact, already have established—the number of particle families: 3. Unfortunately, the number of holes contained in each of the tens of thousands of known Calabi-Yau shapes spans a wide range. Some have 3. But others have 4, 5, 25, and so on—some even have as many as 480 holes. The problem is that at present no one knows how to deduce from the equations of string theory which of the Calabi-Yau shapes constitutes the extra spatial dimensions. If we could find the principle that allows the selection of one Calabi-Yau shape from the numerous possibilities, then indeed a stone from the mountaintop would go tumbling down into the experimentalists' camp. If the particular Calabi-Yau shape singled out by the equations of the theory were to have three holes, we would have found an impressive postdiction from string theory explaining a known feature of the world that is otherwise completely mysterious. But finding the principle for choosing among Calabi-Yau shapes is a problem that as yet remains unsolved. Nevertheless—and this is the important point—we see that string theory provides the potential for answering this basic puzzle of particle physics, and this in itself is substantial progress.

The number of families is but one experimental consequence of the geometrical form of the extra dimensions. Through their effect on possible patterns of string vibrations, other consequences of the extra dimensions include the detailed properties of the force and matter